

100 Days of Azure DevOps

Day 43 Handout — Parameters are the dials. Outputs are the handshake. Hardcoded names are how labs become landfills

5 - Infrastructure as Code · Personal lab only · Educational content · Not a sales pitch

Architecture A — parameters, variables, outputs

Piece	Lives in	Consumed by
parameters	template + @main.parameters.json	az deployment group create -p @file.json
variables	template only (computed)	Not overridden at deploy time
outputs	template outputs: { endpoint: { value: ... } }	Next script / pipeline; query via az deployment group show
uniqueString()	expression in name	Global uniqueness for storage/key vault
nested templates	literacy — extra deployment resource	Skip matryoshka today

Architecture B — what to parameterize

Value	Parameter?	Why
location / sku name	Yes	Second env must not fork the template
secret / key	securestring param or Key Vault ref	Never default a real secret in JSON
apiVersion	No	That is the resource contract, not an env dial
storage account name	Prefix param + uniqueString	Fully hardcoded names collide globally

Architecture C — pitfalls

Pitfall	What you see	Fix
Edited copies of main.json	stprodprod in the wrong region	One template, per-env parameter files
Forgot @ on -p	CLI treats a path as a literal string	-p @main.parameters.json
No outputs	Next step greps the Portal	output the blob endpoint / resourceId
securestring in logs	Parameter value visible in a screenshot	securestring; do not echo; dummy values only

One-liner to remember

Parameters are the dials; hardcoding names is how labs become landfills. Outputs are how the next stack finds the endpoint.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal subscription. RG rg-day43-lab. If Terraform is primary, still do one parameterized deploy — then delete.

- 1 Extend yesterday's template with parameters location and skuName, plus an output storageEndpoint.
- 2 Create infra/main.parameters.json with location=centralindia (or yours) and skuName=Standard_LRS.
- 3 az group create -n rg-day43-lab -l <location>. Deploy: az deployment group create -g rg-day43-lab -f infra/main.json -p @infra/main.parameters.json.
- 4 az deployment group show -g rg-day43-lab -n <name> --query properties.outputs -o jsonc. Confirm the endpoint without opening Portal blades as the source of truth.
- 5 If Terraform is primary: write three lines in docs/arm-params-day43.md and still complete one deploy.
- 6 az group delete -n rg-day43-lab --yes --no-wait.

Parameterized deploy + output query

```
# infra/main.parameters.json
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentParameters.json#",
  "contentVersion": "10.0.0.1",
  "parameters": {
    "location": { "value": "centralindia" },
    "skuName": { "value": "Standard_LRS" }
  }
}

az deployment group create -g rg-day43-lab -n stor43 \
  -f infra/main.json -p @infra/main.parameters.json
az deployment group show -g rg-day43-lab -n stor43 \
  --query properties.outputs.storageEndpoint.value -o tsv
```

Done checklist

- Used -p @parameters.json (the @ matters)
- Output printed from CLI, not from memory
- No real secrets in parameter files
- RG deleted

Tomorrow — Day 44

Bicep fundamentals — same control plane, resource declarations, uniqueString, less brace sport.